

Divergent changes in serum sterols during a strict uncooked vegan diet in patients with rheumatoid arthritis.



The effects of a strict uncooked vegan diet on serum lipid and sterol concentrations were studied in patients with rheumatoid arthritis. The subjects were randomized into a vegan diet group (n 16), who consumed a vegan diet for 2-3 months, or into a control group (n 13), who continued their usual omnivorous diets. Serum total and LDL-cholesterol and -phospholipid concentrations were significantly decreased by the vegan diet. The levels of serum cholestanol and lathosterol also decreased, but serum cholestanol:total cholesterol and lathosterol:total cholesterol did not change. The effect of a vegan diet on serum plant sterols was divergent as the concentration of campesterol decreased while that of sitosterol increased. This effect resulted in a significantly greater sitosterol:campesterol value in the vegan diet group than in the control group (1.48 (SD 0.39) v. 0.72 (SD 0.14); $P < 0.001$). A higher concentration of campesterol compared with sitosterol is normal in omnivorous subjects and can be explained by lower absorption and esterification rates of sitosterol. Our results suggest that a strict uncooked vegan diet changes the relative absorption rates of these sterols and/or their biliary clearance.